



AHEAD OF WHAT'S POSSIBLE™

ADI Design Challenge in Advanced Electronic Circuits ECSE 4050, ECSE 6050

Spring 2026

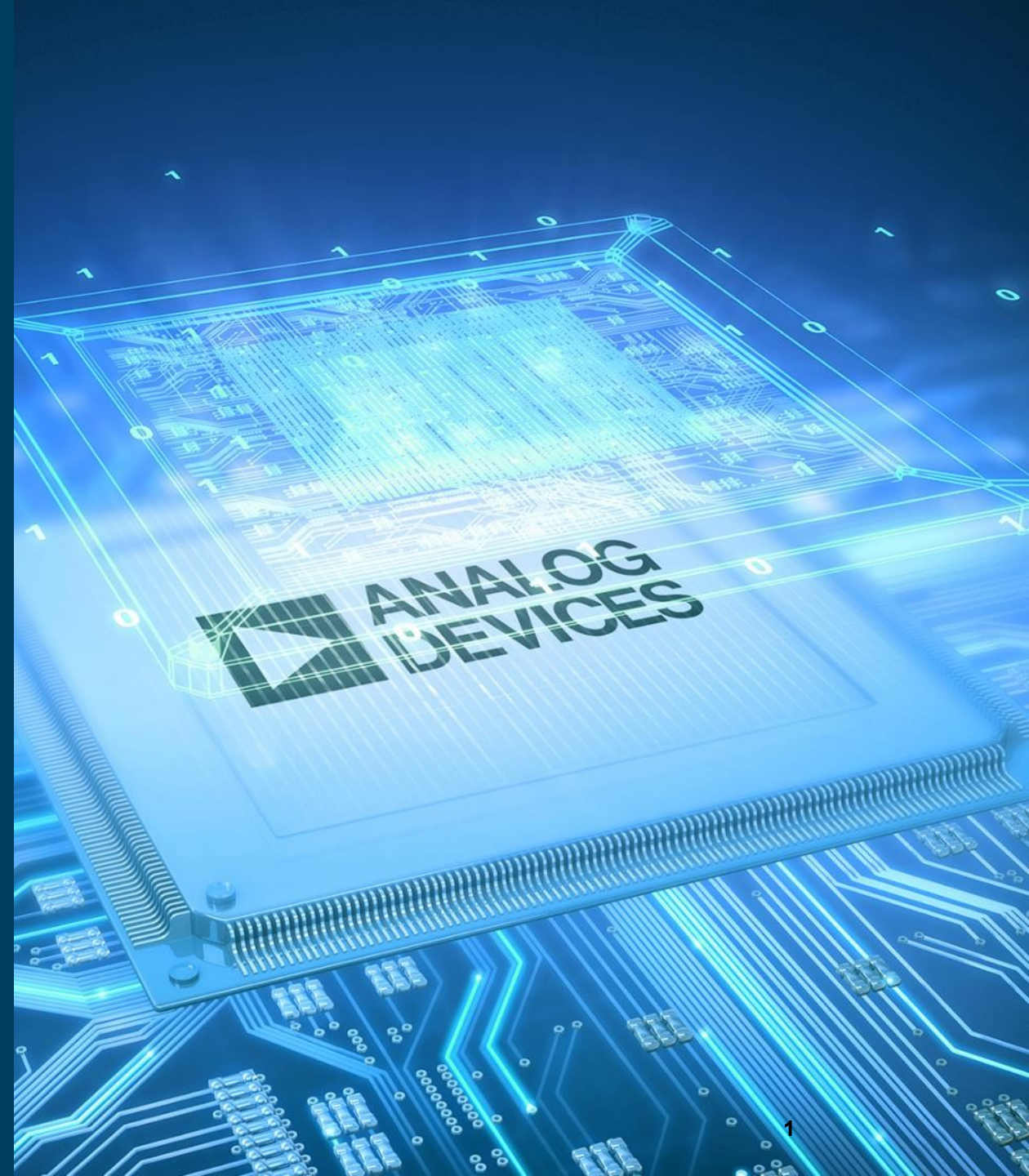
Feb 26th 2026 Kick-off

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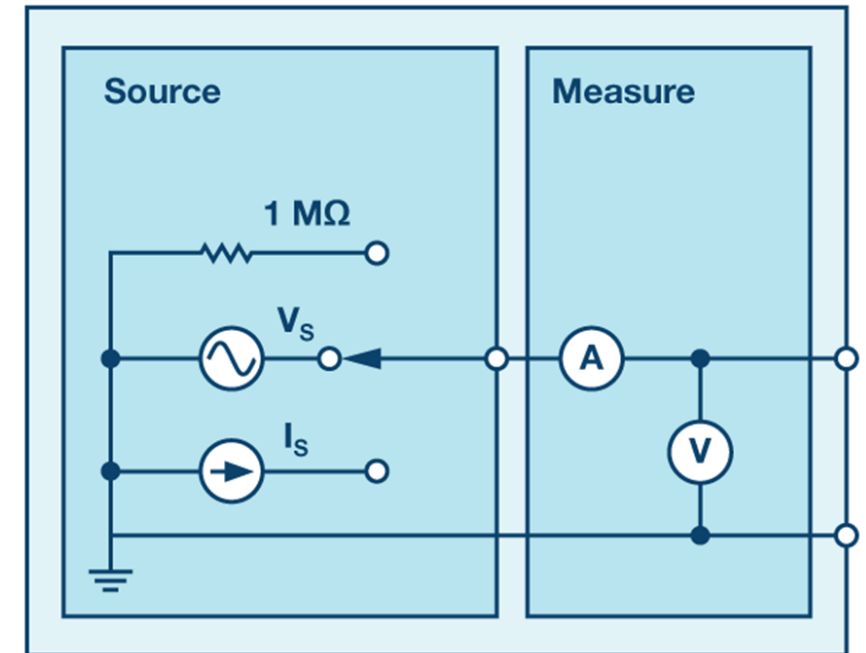
Design Challenge goals

- Part 1
 - The Design of a configurable Voltage / Current source (as part of a source measure unit or SMU)
 - Based on integrated circuit amplifiers
 - Verify spice models and test bench through simulation
 - Optimize the frequency response / stability while driving various R / C loads
 - Simulate / analyze frequency and time domain response in both output modes
- Part 2 (optional for 4050 as extra bonus credit, required for 6050)
 - Simulate / analyze frequency and time domain response with inductive loads (L/R)
 - Select alternative ICs

Design Challenge Background

- ▶ The source measure unit (SMU), is an electronic instrument that provides both sourcing and measuring function on the same pin. It can precisely force voltage or current and simultaneously measure precise voltage and/or current.
- ▶ The SMU integrates the capabilities of a power supply or function generator, a digital multi-meter (DMM) or oscilloscope, a current source, and an electronic load into a single, tightly synchronized instrument.
- ▶ A SMU consists of:
 - ❑ A Voltage Source
 - ❑ A Current Source
 - ❑ A Volt Meter
 - ❑ An Ammeter

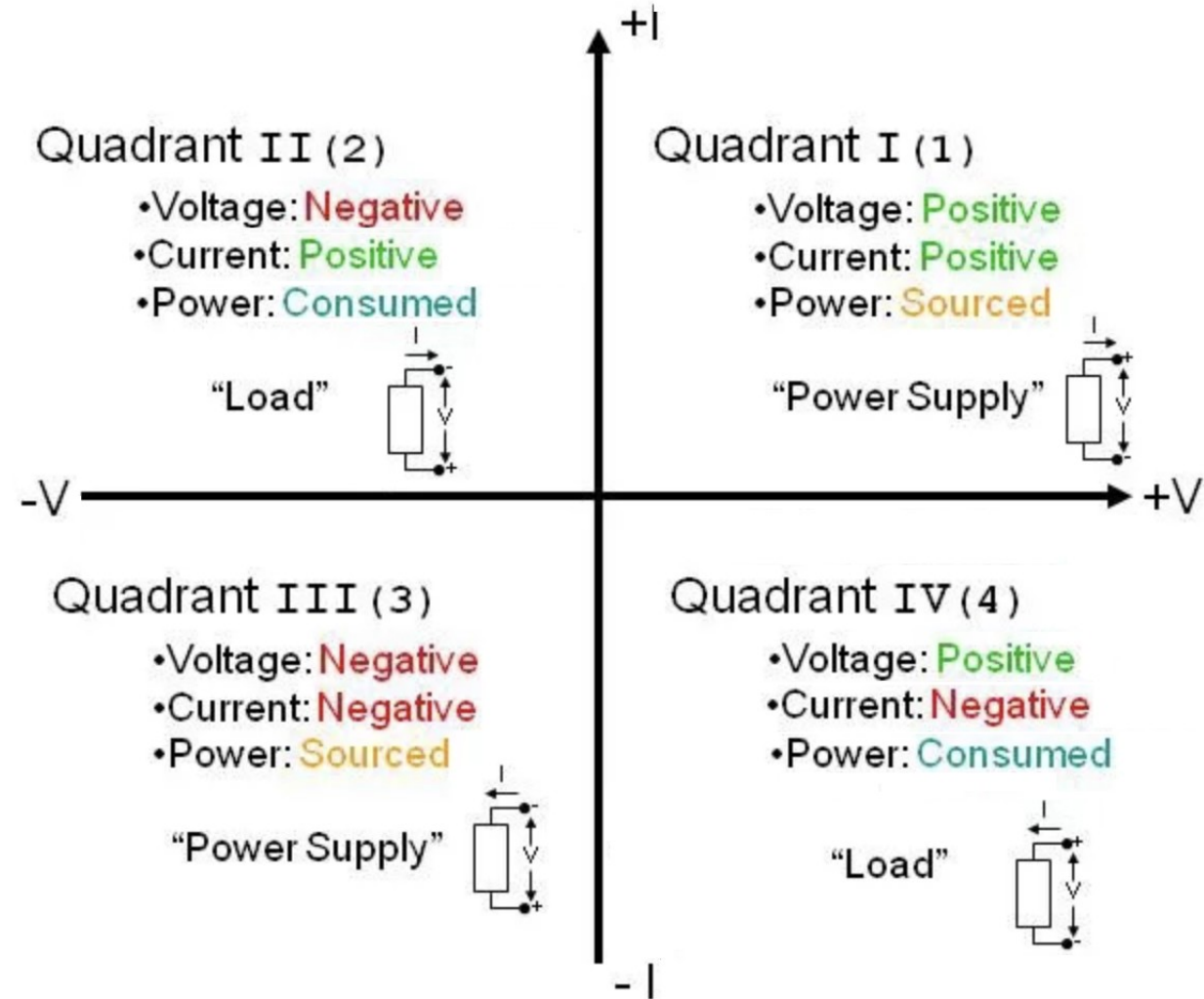
The SMU is a critical part of many laboratory bench and production test systems.



More Background

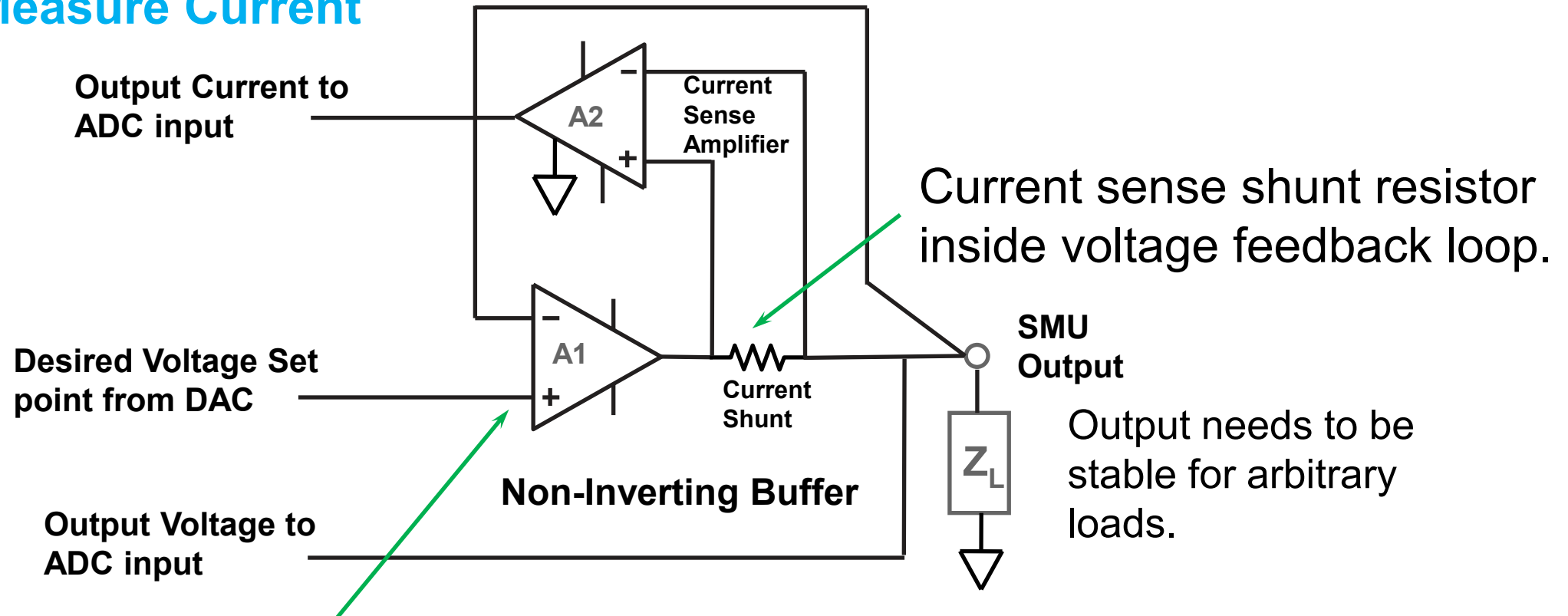
► Four Quadrant operation

Sources and Loads can be considered to operate in one of four quadrants



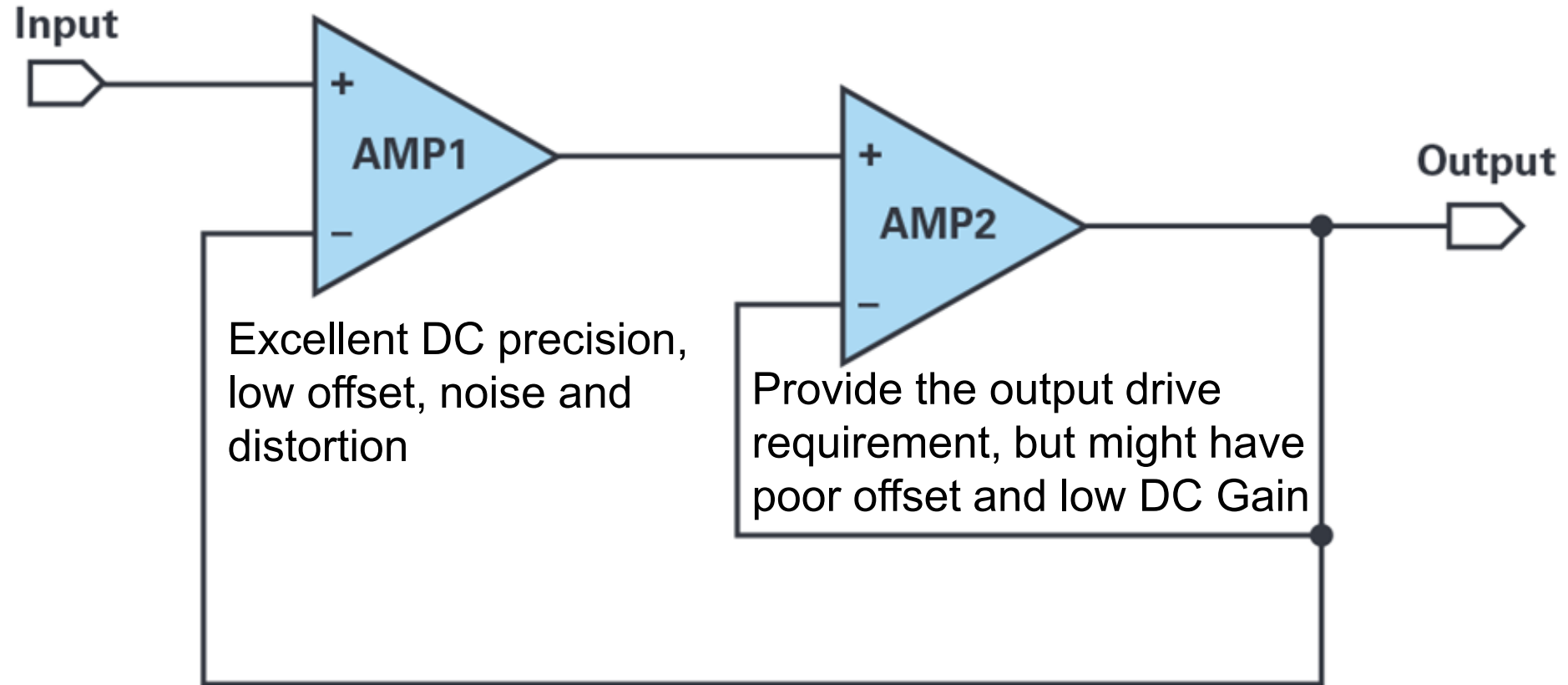
SMU as A Voltage Source

Source Voltage Measure Current



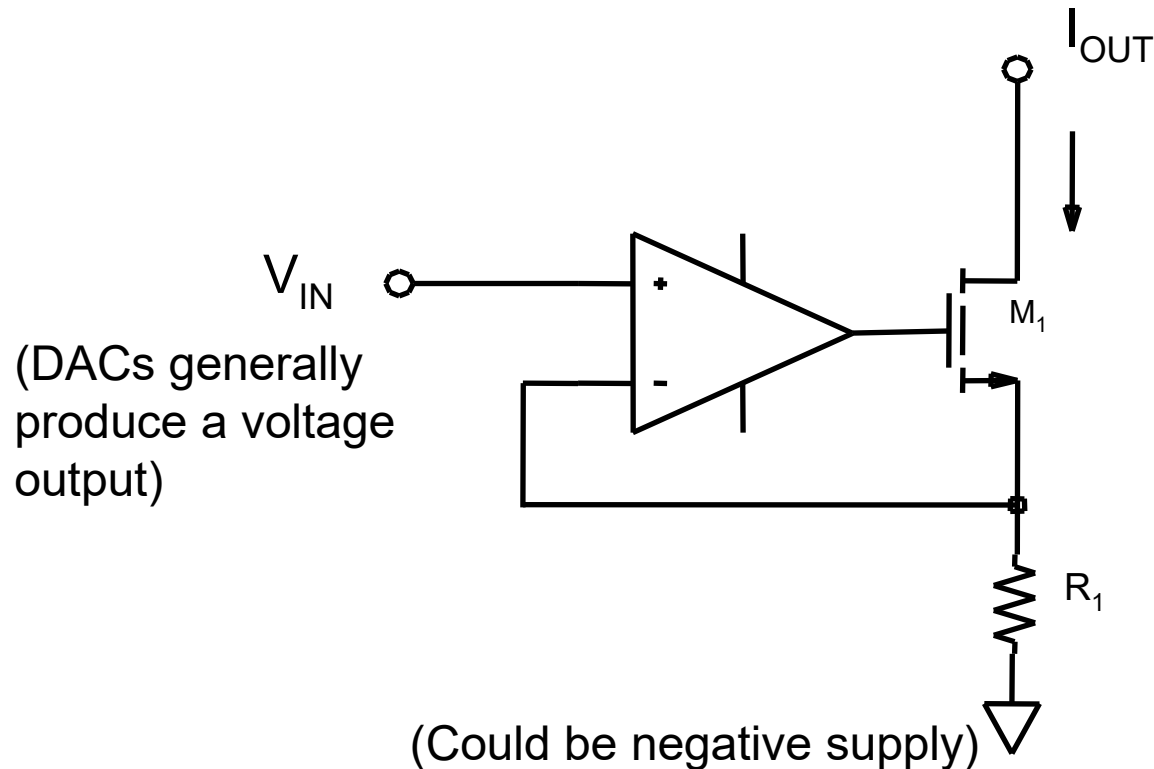
Buffer A1 needs to provide the specified output drive current over the specified voltage range while maintaining the required accuracy i.e. low offset, good linearity.

The Composite Amplifier: High Output Drive Capability with Precision



A composite amplifier is an arrangement of two individual amplifiers configured in such a way as to realize the benefits of each individual amplifier while diminishing the drawbacks of each amplifier.

Precision Voltage to Current Converter



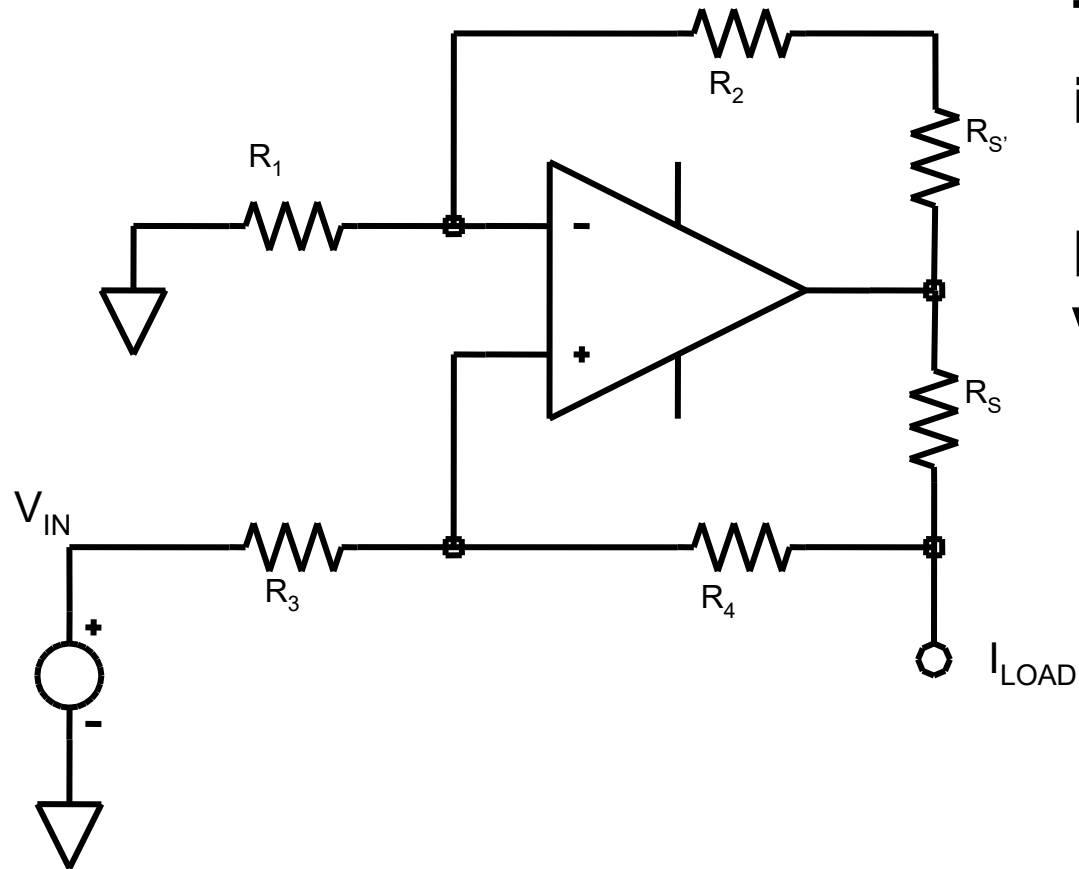
Has Draw Backs:

Only Sinks Current i.e. operates in Quadrants III and IV

Can be flipped over using a P-Type device to Source Current for operation in Quadrants I and II

But stability is generally not affected by changes in output load

Active Voltage to Current Converter, Howland current pump

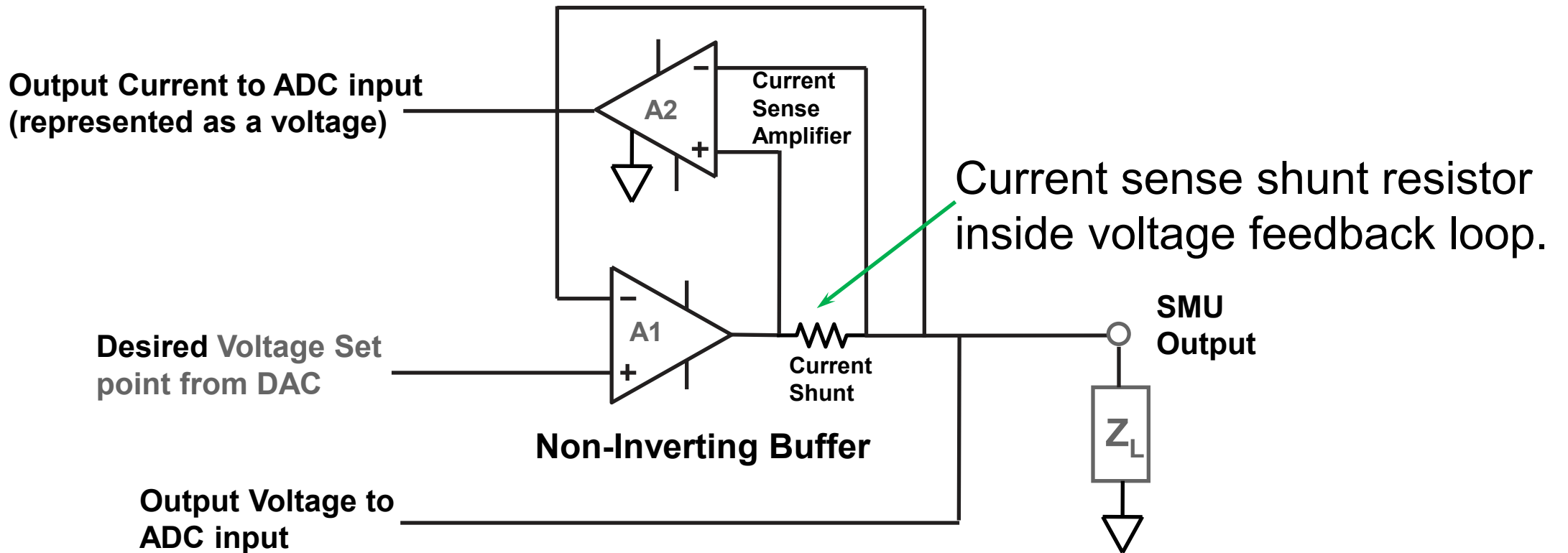


**This circuit can Source and Sink current.
i.e. operate in all four quadrants.**

**But is not easily re-configured as a
Voltage source.**

SMU as Voltage Source

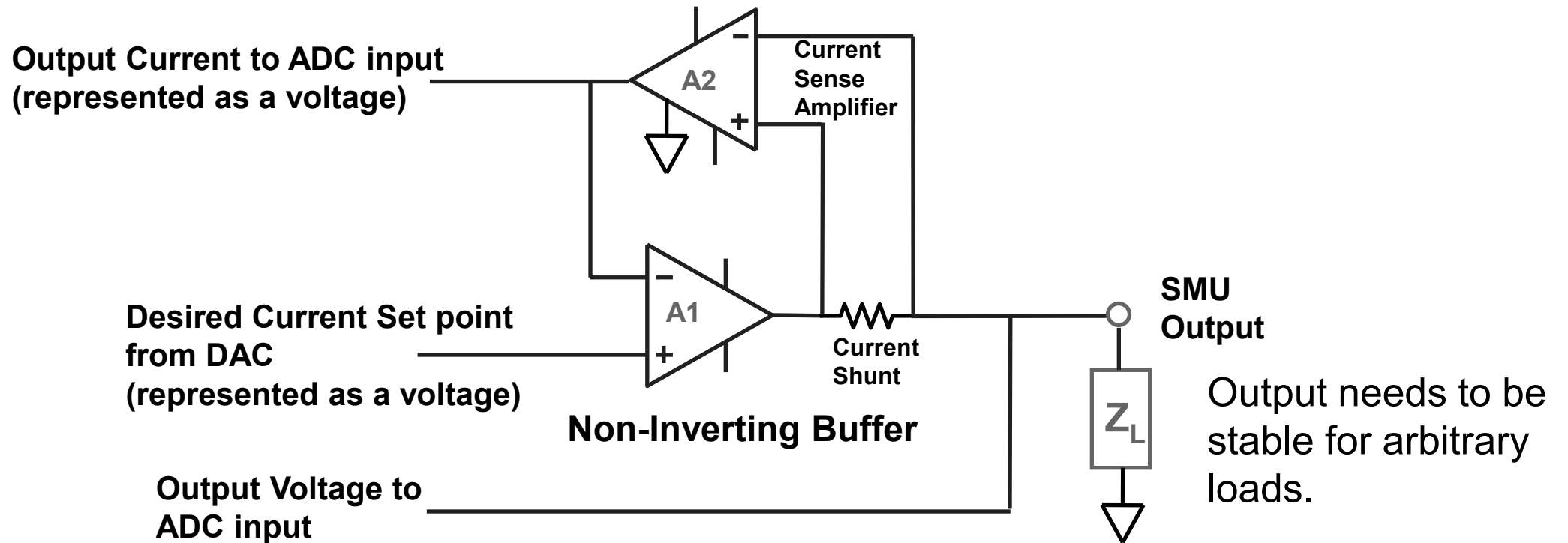
Source Voltage Measure Current



SMU as Current Source

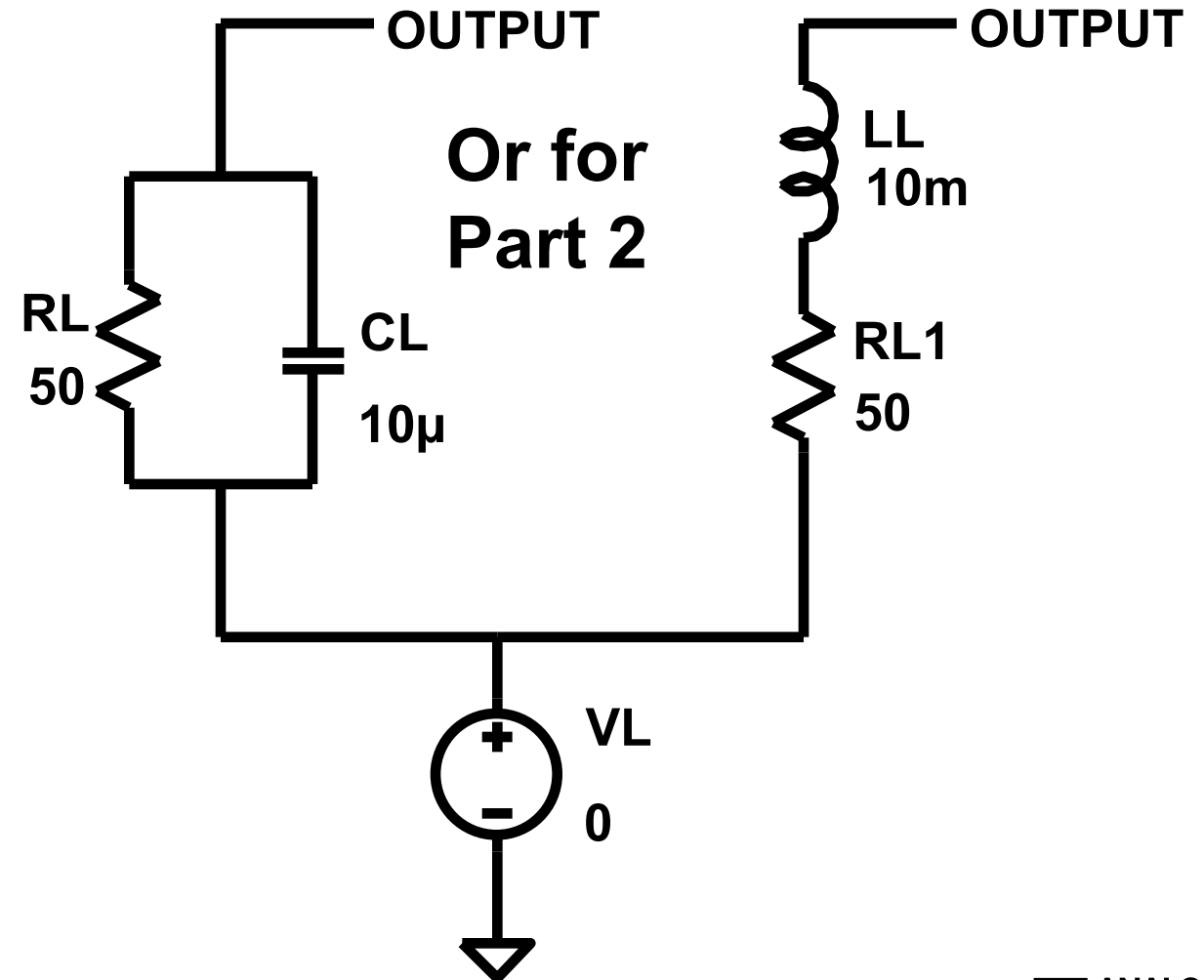
Source Current
Measure Voltage

Current fed back as a voltage.



Modeling the Load with an AC Thevenin Equivalent

Any combination of (DC and AC) sources and impedances with two terminals can be replaced by a single voltage source V_L and a single series or parallel impedance Z_L . The value of V_L is the open circuit voltage at the terminals. In this case, the impedance evaluation involves a series-parallel combination of R, C and L.



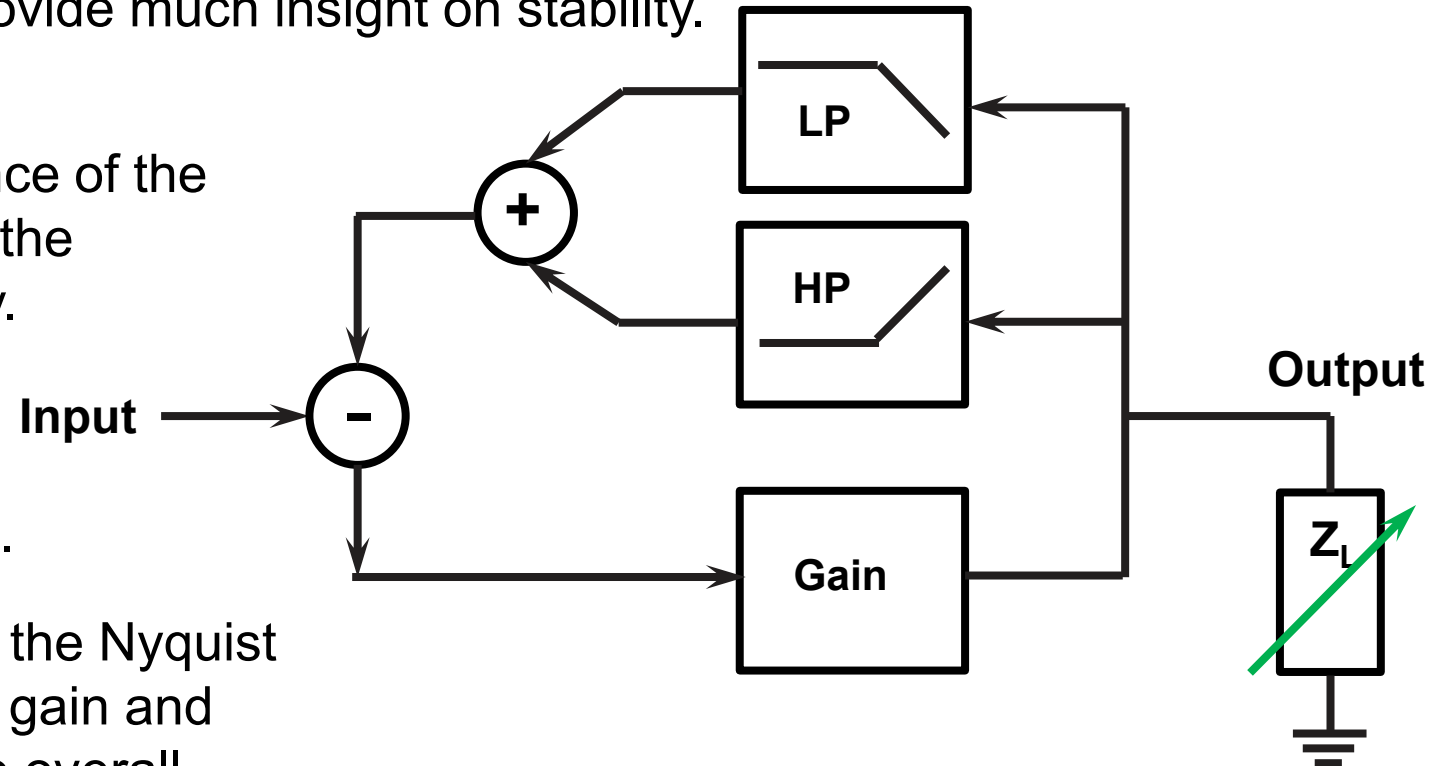
Notes on Negative Feedback and Stability

Simple Bode Plots measure the input to output gain and phase response of the closed loop system. They do not always provide much insight on stability.

Finite Input and Output impedance of the various blocks in the loop affect the frequency response and stability.

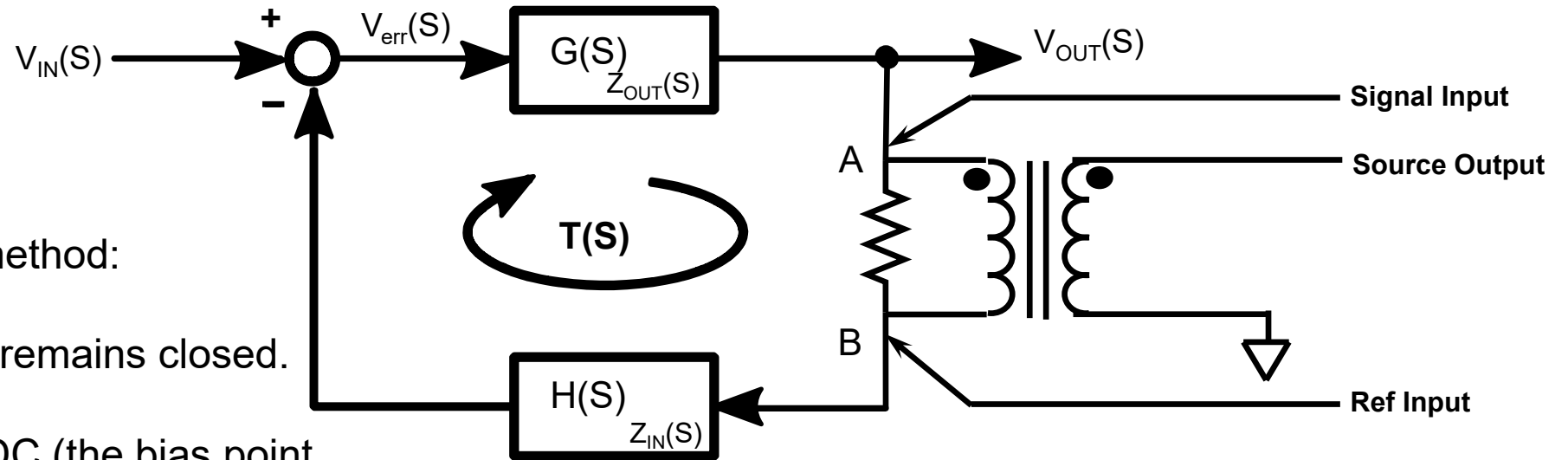
We need to break open the loop.

With the loop gain we can apply the Nyquist stability criterion to measure the gain and phase margin and determine the overall stability of the closed loop system.



Voltage Injection Method (Middlebrook technique)

One technique to measure the loop gain in negative feedback systems is the voltage injection method.



Advantage of Middlebrook's method:

It acts as if the feedback path remains closed.

However, this is true only for DC (the bias point remains unchanged).

The floating AC source decouples both nodes (A and B) providing two AC voltages determined by the loop gain.

These voltages, of course, are identical to the values resulting from actually opening the loop.

In the real world (the Lab) a low value resistor is inserted in the feedback loop at the injection-point. An Injection transformer drives (inserts) a floating voltage.

In simulation we can use an ideal floating voltage source.

Determining Boundary Conditions

Givens:

1. Output Voltage amplitude is to be 0 to +/- 2.5 V or a step size up to 5 V (as determined by amplifier power supply)
2. Sinewave Frequency from 0 to 100 KHz
3. Amplifier (AD8018) can supply up to a peak current of 400 mA
4. The possible load capacitance will be from 0 to 10 μ F

When driving a capacitor the Frequency and Amplitude can be used to calculate the peak current in the capacitor.

There are three free variables: Frequency, Amplitude and Capacitance. This forms a three dimensional surface as the boundary condition.

Calculate the maximum Frequency for the maximum +/- 2.5 V amplitude and maximum capacitor value based on the given 400 mA peak current.

Before starting the frequency compensation design

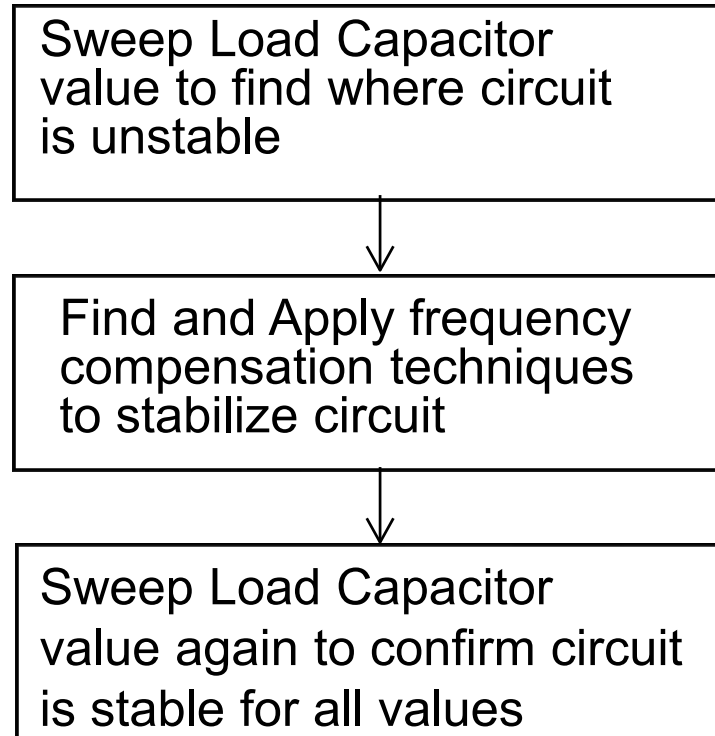
As a first step we highly recommend that you verify that the Spice models and the test bench schematic is working before starting the actual frequency compensation circuit design (with zero load capacitance). Get familiar with getting the circuit to DC converge.

It might be helpful to write a short (2-3 page) report documenting your verification results before moving to the next step.

A model verification time domain test bench LTSpice schematic file (or files) will be provided to get you started. The first provided example uses an ideal current to voltage converter that does not model any time or phase delay. A simple R/C delay can be added to model non-ideal behavior.

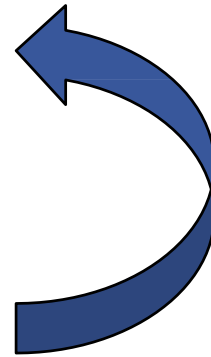
Once you are familiar with the simulation test bench(s) determine the range of load capacitances where circuit is and is not stable. How much capacitance and at what frequencies?

Design flow



Start: Stabilize the circuit in voltage output mode (i.e. the composite amplifier)

Then: Move to the Current output mode with the current sense amplifier included

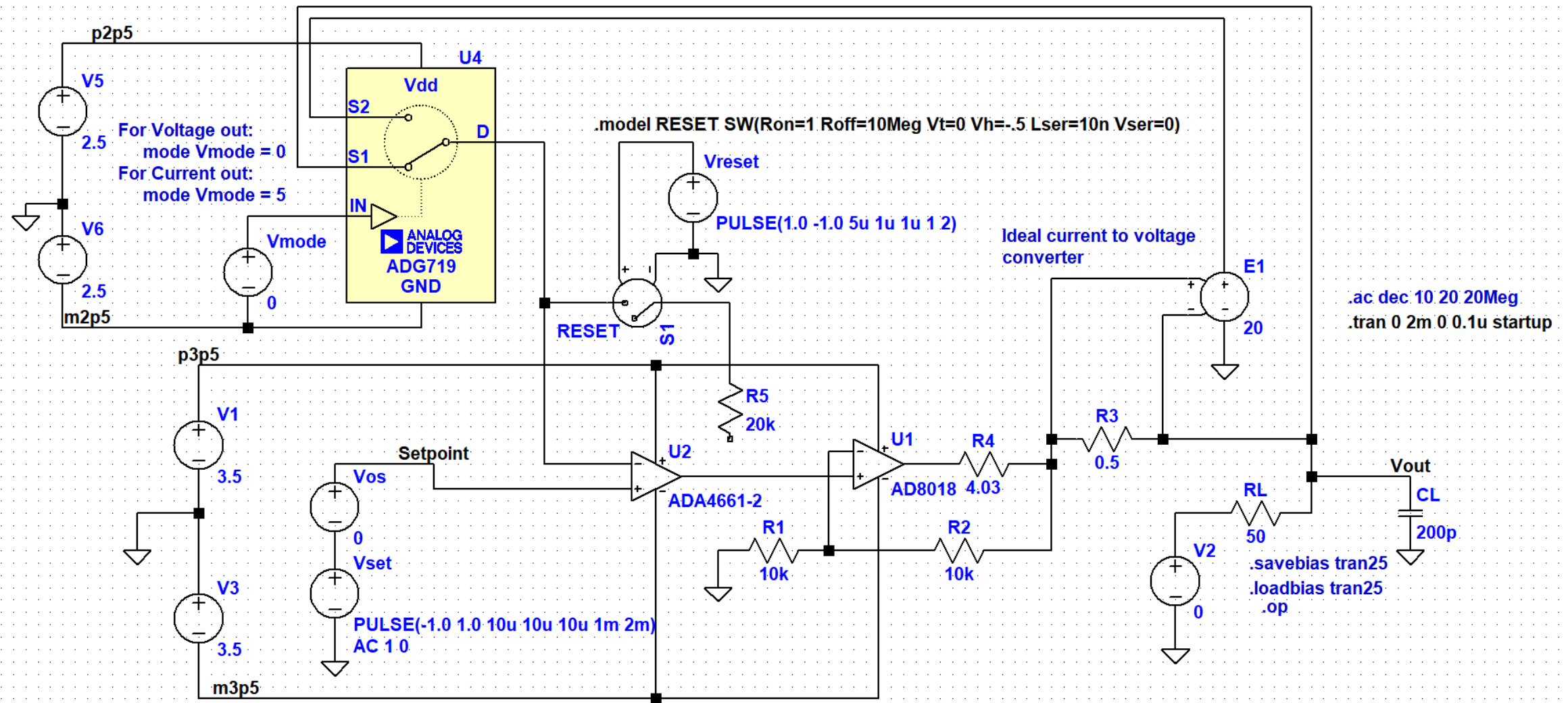


No? Go back and try another technique

Circuit Simulation Software

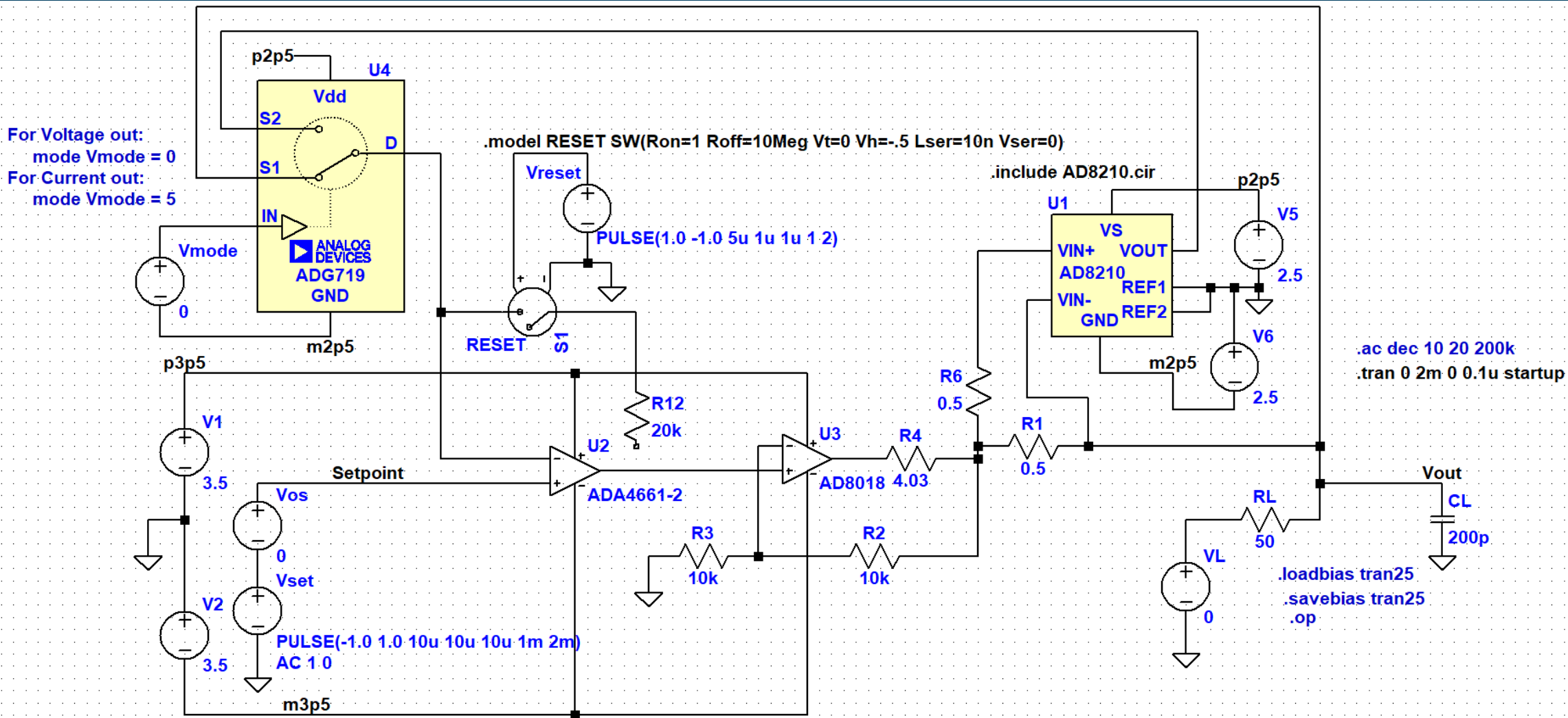
- ◆ **LTSpice:** (<https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>)
 - ◆ Down loads: Windows 11 **Current Version 26.0**, MacOS Version 17
 - (Linux use Wine64 and install Windows version)
- ◆ **For this design problem we will supply example test bench schematics with all needed component symbols and models to get you started.**

Basic Source Operation (test bench simulation 1)



This Simulation contains NO Frequency compensation and is NOT stable!

Adding Real I to V Amplifier (test bench simulation 2)



This Simulation contains NO Frequency compensation and is NOT stable!

Design Challenge details

- ▶ Source specification target goals:
 - It must operate over a -2.5V to +2.5V analog voltage output range.
 - It must operate over a -200mA to +200mA analog current output range.
 - Power supply voltages to be no larger than +/- 3.5V.
 - The target range for the Load Capacitor is between 400 pF Min and 10 uF Max
 - For Part 2, The target range for the Load Inductance is up to 100 mH.
 - **The Best design will be one that maximizes the following:**
 - Bandwidth / settling time
 - Widest range of loads

Challenge details

- ▶ Follow the instructions/guidelines provided
- ▶ Design must consist of **only AD or LT or MAX** part numbers.
- ▶ In your write-up please include the following at a minimum:
 - Any Hand calculations, analysis
 - Clearly annotated schematic diagrams
 - Bode sweeps and transient simulation plots (basic operation, ...)
 - Any other relevant simulation plots (from LTSpice / MATLAB etc.)
 - Explanation of circuit topologies used (must be provided in detail)
 - Please **reference** all application notes, datasheets and text books you used

Challenge details

- ▶ **Deadline for all submissions : Thursday, April 16th(?) 2026 11:59 pm EDT**
 - soft copy to be emailed to TA or Prof. Wilt (not ADI)
 - Prefer PDF copies
 - Please name your files: lastname1_lastname2_spring2026.pdf
- ▶ Expect to give a 5-10 min presentation of your design in class (April 27th)
 - Depending on size of class the best 5 or 6 papers may be preselected shortly after the deadline
- ▶ Be sure to indicate which section, -4050 or -6050 you are in.
- ▶ The winner will be selected based on a variety of criteria:
 - Best design (meets challenge criteria, most innovative, adventurous, lowest cost etc ...)
 - Quality of write-up (clarity of all schematic diagrams and simulation plots)
 - Hand calculations to support simulation results
 - Demonstrating an understanding of the fundamental concepts
 - Completeness of report

Helpful Links:

- ▶ LTspice: <https://www.analog.com/en/resources/design-tools-and-calculators/ltspice-simulator.html>
 - Downloads: Windows 10 **Current Version 26**, MacOS Version 17
- ▶ ADI Spice macro models: <https://www.analog.com/en/resources/simulation-models/spice-models.html>
- ▶ Composite Amplifiers: <https://www.analog.com/en/analog-dialogue/articles/composite-amplifiers-high-output-drive-capability-with-precision.html>
- ▶ **Current Feedback (CFB) Op Amps**: <https://www.analog.com/media/en/training-seminars/tutorials/MT-034.pdf>
- ▶ **Effects of Feedback Capacitance on VFB and CFB Op Amps**: <https://www.analog.com/media/en/training-seminars/tutorials/MT-058.pdf>
- ▶ **Current Feedback Amplifier “Do’s and Don’ts”**: <https://www.analog.com/media/en/reference-design-documentation/design-notes/dn46fa.pdf>
- ▶ Practical Techniques to Avoid Instability Due to Capacitive Loading: <https://www.analog.com/en/analog-dialogue/articles/techniques-to-avoid-instability-capacitive-loading.html>
- ▶ Loop gain measurements: <https://www.edn.com/electronics-blogs/analog-bytes/4434609/Loop-gain-measurements->
- ▶ Op-Amp Stability Basics: <https://ez.analog.com/precision-technology-signal-chains/w/kwik-circuits/32237/op-amp-stability-basics-part-1>

Contest details

- ▶ Judges will select the best 3 designs from each section after the in class presentations...
- ▶ ECSE-4050
 - 1st prize: \$400
 - 2nd prize: \$200
 - 3rd prize: \$150
- ▶ ECSE-6050
 - 1st prize: \$400
 - 2nd prize: \$200
 - 3rd prize: \$150
- ▶ Prize distribution based on number of papers submitted and subject to adjustment in case of ties.
- ▶ Decision of the judges is final!

Additional Slides

Aid to DC convergence

- ▶ Sometimes starting Trans Sims with Setpoint at 0.0 V converges while +/- 0.5 V or greater does not. Adding startup to .trans might help. Setting initial condition of load Cap might help.
- ▶ Steps to Follow:
 - Enable RESET switch connecting resistor at output of U2
 - Turn on .op and .savebias directives
 - Turn off .ac and/or .tran and .loadbias directives (change to a comment)
 - Run a simulation
 - You should now have saved operating point values in a file
 - Disable RESET switch disconnecting resistor
 - Turn on .ac and/or .tran and .loadbias directives
 - Turn off .op and .savebias directives
 - Run simulation (which ever is turned on , Frequency sweep or time etc.)
- ▶ May need to repeat these steps if schematic is changed i.e. you add or remove components